

Databricks Generative AI Practice Questions

Questions 30-56 | Full options and verified answer key

This study guide reproduces the questions and answer choices submitted in this chat. The correct choice is highlighted under each question for quick review.

QUICK ANSWER KEY

30: A	31: A	32: B	33: C	34: A	35: D
36: D	37: A	38: C	39: A	40: C	41: C
42: C	43: A	44: A	45: B	46: A	47: B
48: C	49: D	50: B	51: C	52: C	53: D
54: A and D	55: A	56: A			

For Question 54, select two answers: A and D.

QUESTION 30

A Generative AI Engineer has created a RAG application which can help employees retrieve answers from an internal knowledge base, such as Confluence pages or Google Drive. The prototype application is now working with some positive feedback from internal company testers. Now the Generative AI Engineer wants to formally evaluate the system's performance and understand where to focus their efforts to further improve the system.

How should the Generative AI Engineer evaluate the system?

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|-----------|--|----------------|
| A. | Create a dataset that can test the retrieval and generation components of the system separately. Use MLflow's built-in evaluation metrics to perform the evaluation on the retrieval and generation components. | CORRECT |
| B. | Benchmark multiple LLMs with the same data and pick the best LLM for the job. | |
| C. | Use an LLM-as-a-judge to evaluate the quality of the final answers generated. | |
| D. | Use cosine similarity score to comprehensively evaluate the quality of the final generated answers. | |

Correct answer: A. Create a dataset that can test the retrieval and generation components of the system separately. Use MLflow's built-in evaluation metrics to perform the evaluation on the retrieval and generation components.

QUESTION 31

What is an effective method to preprocess prompts using custom code before sending them to an LLM?

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| A. | Write an MLflow PyFunc model that has a separate function to process the prompts. | CORRECT |
| B. | Directly modify the LLM's internal architecture to include preprocessing steps. | |
| C. | Rather than preprocessing prompts, it is more effective to postprocess the LLM outputs to align the outputs to desired outcomes. | |
| D. | It is better not to introduce custom code to preprocess prompts, as the LLM has not been trained with examples of the preprocessed prompts. | |

Correct answer: A. Write an MLflow PyFunc model that has a separate function to process the prompts.

QUESTION 32

A Generative AI Engineer has been asked to build an LLM-based question-answering application. The application should take into account new documents that are frequently published. The engineer wants to build this application with the least cost and least development effort and have it operate at the lowest cost possible.

Which combination of chaining components and configuration meets these requirements?

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| A. | For the application, a prompt, an agent, and a fine-tuned LLM are required. The agent is used by the LLM to retrieve relevant content that is inserted into the prompt, which is given to the LLM to generate answers. | |
| B. | For the application, a prompt, a retriever, and an LLM are required. The retriever output is inserted into the prompt, which is given to the LLM to generate answers. | CORRECT |
| C. | The LLM needs to be frequently fine-tuned with the new documents in order to provide the most up-to-date answers. | |
| D. | For the question-answering application, prompt engineering and an LLM are required to generate answers. | |

Correct answer: B. For the application, a prompt, a retriever, and an LLM are required. The retriever output is inserted into the prompt, which is given to the LLM to generate answers.

QUESTION 33

A Generative AI Engineer has successfully ingested unstructured documents and chunked them by document sections. They would like to store the chunks in a Vector Search index. The current format of the dataframe has two columns: (i) original document file name and (ii) an array of text chunks for each document.

What is the most performant way to store this dataframe?

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| A. | Store each chunk as an independent JSON file in a Unity Catalog Volume. For each JSON file, the key is the document section name and the value is the array of text chunks for that section. | |
| B. | Split the data into train and test sets, create a unique identifier for each document, then save to a Delta table. | |
| C. | Flatten the dataframe to one chunk per row, create a unique identifier for each row, and save to a Delta table. | CORRECT |
| D. | First, create a unique identifier for each document, then save to a Delta table. | |

Correct answer: C. Flatten the dataframe to one chunk per row, create a unique identifier for each row, and save to a Delta table.

QUESTION 34

A Generative AI Engineer is implementing a supervisor agent and two specialist agents in Databricks: a Sales Analyst for revenue questions and an HR Analyst for stock questions. Each specialist must retrieve data only from its own governed domain, and the engineer wants to preserve that separation using Databricks-native data access for each agent rather than building custom retrieval logic.

What should the engineer implement?

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| A. | Create two separate Genie Spaces for Sales and HR, each scoped to its own governed datasets, and have each specialist agent call the appropriate Space through the API. | CORRECT |
| B. | Create a single Genie Space over both domains and rely on the supervisor agent to route only sales questions to the Sales Analyst and HR questions to the HR Analyst. | |
| C. | Create separate Knowledge Assistants for Sales and HR and have each specialist retrieve from the corresponding assistant. | |
| D. | Create a shared Genie Space over both domains, but use distinct service principals and Unity Catalog grants for each specialist agent's API access. | |

Correct answer: A. Create two separate Genie Spaces for Sales and HR, each scoped to its own governed datasets, and have each specialist agent call the appropriate Space through the API.

QUESTION 35

A Generative AI Engineer is deploying a conversational agent that provides financial analysis insights to internal analysts at a company. The company has standardized on Microsoft 365 for all internal communications and collaboration. The analysts are distributed across multiple time zones and primarily work within Microsoft Teams channels where they already discuss financial data and share reports. The agent needs to integrate with the company's existing Azure Active Directory for authentication and must respect the company's data governance policies. Users should be able to interact with the agent within their existing workflow without switching applications.

Which interface should the engineer implement for this agent deployment?

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| A. | Databricks Apps to create a custom web application that analysts can access through their browsers with SSO authentication. | |
| B. | Slack integration to enable the agent in a dedicated workspace for the financial analysis team. | |
| C. | A REST API endpoint with a custom frontend application hosted on the company's internal portal. | |
| D. | Microsoft Teams integration to deploy the agent as a bot within Teams channels where analysts already collaborate. | CORRECT |

Correct answer: D. Microsoft Teams integration to deploy the agent as a bot within Teams channels where analysts already collaborate.

QUESTION 36

A small and cost-conscious startup in the cancer research field wants to build a RAG application using Foundation Model APIs.

Which strategy would allow the startup to build a good-quality RAG application while being cost-conscious and able to cater to customer needs?

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|-----------|---|----------------|
| A. | Limit the number of queries a customer can send per day. | |
| B. | Use the largest LLM possible because that gives the best performance for any general queries. | |
| C. | Limit the number of relevant documents available for the RAG application to retrieve from. | |
| D. | Pick a smaller LLM that is domain-specific. | CORRECT |

Correct answer: D. Pick a smaller LLM that is domain-specific.

QUESTION 37

A Generative AI Engineer at a legal firm is designing a RAG system to analyze historical legal case precedents. The system needs to process millions of court opinions and legal documents, already organized by time and topic, to track how interpretations of specific laws have evolved over time. All of these documents are in plain text. The engineer needs to choose a chunking method that would most effectively preserve continuity and the temporal nature of the cases.

Which method do they choose?

A.	Implement windowed summarization with overlapping chunks.	CORRECT
B.	Implement a hierarchical tree structure, like RAPTOR, to group similar legal concepts.	
C.	Implement paragraph-level embeddings with each chunk.	
D.	Implement sentence-level embeddings with each chunk tagged with the time to enable metadata filtering.	

Correct answer: A. Implement windowed summarization with overlapping chunks.

QUESTION 38

A Generative AI Engineer just deployed an LLM application at a digital marketing company that assists with answering customer service inquiries.

Which metric should they monitor for their customer service LLM application in production?

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| A. | Hugging Face Leaderboard values for the base LLM. | |
| B. | Final perplexity scores for the training of the model. | |
| C. | Number of customer inquiries processed per unit of time. | CORRECT |
| D. | Energy usage per query. | |

Correct answer: C. Number of customer inquiries processed per unit of time.

QUESTION 39

A Generative AI Engineer is managing prompt templates using MLflow v3.x for a document summarization pipeline. A regulatory audit requires the team to demonstrate exactly which prompt version was used to generate outputs on a specific date three months ago, including the exact prompt text and any variables used at that time.

Which combination of MLflow v3.x capabilities allows the engineer to satisfy this audit requirement?

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| A. | MLflow Prompt Registry version history and logged runs that reference the prompt name and version used during inference. | CORRECT |
| B. | MLflow experiment tags and an automatically scripted changelog stored in a Databricks notebook. | |
| C. | MLflow Model Registry webhooks and a downstream audit log stored in an external database. | |
| D. | MLflow autologging and Delta Lake time travel on the inference table. | |

Correct answer: A. MLflow Prompt Registry version history and logged runs that reference the prompt name and version used during inference.

QUESTION 40

A Generative AI Engineer is creating an LLM-based application. The documents for its retriever have been chunked to a maximum of 512 tokens each. The Generative AI Engineer knows that cost and latency are more important than quality for this application. They have several context length levels to choose from.

Which will fulfill their need?

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| A. | Context length 32768; smallest model is 14 GB and embedding dimension is 4096. | |
| B. | Context length 2048; smallest model is 11 GB and embedding dimension is 2560. | |
| C. | Context length 512; smallest model is 0.13 GB and embedding dimension is 384. | CORRECT |
| D. | Context length 514; smallest model is 0.44 GB and embedding dimension is 768. | |

Correct answer: C. Context length 512; smallest model is 0.13 GB and embedding dimension is 384.

QUESTION 41

When developing an LLM application, it is crucial to ensure that the data used for training the model complies with licensing requirements to avoid legal risks.

Which action is NOT appropriate to avoid legal risks?

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| A. | Reach out to the data curators directly before you have started using the trained model to let them know. | |
| B. | Only use data explicitly labeled with an open license and ensure the license terms are followed. | |
| C. | Reach out to the data curators directly after you have started using the trained model to let them know. | CORRECT |
| D. | Use any available data you personally created which is completely original and you can decide what license to use. | |

Correct answer: C. Reach out to the data curators directly after you have started using the trained model to let them know.

QUESTION 42

A Generative AI Engineer is building a multi-turn chat app with LangGraph on Databricks. The app must persist chat history (messages, roles, timestamps, session IDs) for many concurrent users, support SQL queries, and stay governed in Unity Catalog. The engineer also wants ACID guarantees, low-latency reads/writes, and an easy way to sync chat data into Delta tables for analytics and model training.

Which approach fits these requirements?

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| A. | Write each turn from a custom LangGraph node directly into a Delta table with Spark append, then query history via Spark SQL on every request. | |
| B. | Store conversation history in MLflow runs and retrieve it via the MLflow Tracking API inside LangGraph nodes. | |
| C. | Use Lakebase with a chat_history table wired to a Postgres-backed LangGraph checkpoint/memory component and enable sync from Lakebase into Delta tables. | CORRECT |
| D. | Use a custom in-memory LangGraph state store running on the Databricks cluster driver, and periodically snapshot state to JSON files in DBFS. | |

Correct answer: C. Use Lakebase with a chat_history table wired to a Postgres-backed LangGraph checkpoint/memory component and enable sync from Lakebase into Delta tables.

QUESTION 43

A company has a typical RAG-enabled, customer-facing chatbot on its website.

Select the correct sequence of components a user's questions will go through before the final output is returned.

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| A. | 1. Embedding model; 2. Vector search; 3. Context-augmented prompt; 4. Response-generating LLM. | CORRECT |
| B. | 1. Context-augmented prompt; 2. Vector search; 3. Embedding model; 4. Response-generating LLM. | |
| C. | 1. Response-generating LLM; 2. Vector search; 3. Context-augmented prompt; 4. Response-generating LLM. | |
| D. | 1. Response-generating LLM; 2. Context-augmented prompt; 3. Vector search; 4. Embedding model. | |

Correct answer: A. 1. Embedding model; 2. Vector search; 3. Context-augmented prompt; 4. Response-generating LLM.

QUESTION 44

A company selling gourmet mushroom growing supplies has a script that runs once per day to scrape various social media platforms for posts that mention its name. The scraped text data is loaded into a Delta table each night for a downstream processing task that summarizes each post and its sentiment for internal use. Given the small size of the company, they only get a couple hundred posts per day.

Which solution best optimizes for cost and ease of implementation?

A.	Schedule a nightly SQL query with <code>ai_query()</code> calling a Pay-Per-Token endpoint.	CORRECT
B.	Schedule a nightly job to call OpenAI's batch inference API, save the results, and terminate the cluster.	
C.	Schedule a nightly job that runs a notebook to download the model locally, process the records, and terminate the cluster.	
D.	Schedule a nightly SQL query with <code>ai_query()</code> calling a Provisioned Throughput endpoint.	

Correct answer: A. Schedule a nightly SQL query with `ai_query()` calling a Pay-Per-Token endpoint.

QUESTION 45

A Generative AI Engineer is designing a chatbot for a gaming company that aims to engage users on its platform while its users play online video games.

Which metric would help them increase user engagement and retention for their platform?

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| A. | Lack of relevance. | |
| B. | Diversity of responses. | CORRECT |
| C. | Randomness. | |
| D. | Repetition of responses. | |

Correct answer: B. Diversity of responses.

QUESTION 46

A Generative AI Engineer is tasked with improving the RAG quality by addressing its inflammatory outputs.

Which action would be most effective in mitigating the problem of offensive text outputs?

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| A. | Curate upstream data properly, including manual review before it is fed into the RAG system. | CORRECT |
| B. | Increase the frequency of upstream data updates. | |
| C. | Restrict access to the data sources to a limited number of users. | |
| D. | Inform the user of the expected RAG behavior. | |

Correct answer: A. Curate upstream data properly, including manual review before it is fed into the RAG system.

QUESTION 47

A Generative AI Engineer is developing a chatbot designed to assist users with insurance-related queries. The chatbot is built on a large language model (LLM) and is conversational. However, to maintain the chatbot's focus and to comply with company policy, it must not provide responses to questions about politics. Instead, when presented with political inquiries, the chatbot should respond with a standard message: 'Sorry, I cannot answer that. I am a chatbot that can only answer questions around insurance.'

Which framework type should be implemented to solve this?

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|-----------|------------------------------|----------------|
| A. | Security Guardrail. | |
| B. | Contextual Guardrail. | CORRECT |
| C. | Compliance Guardrail. | |
| D. | Safety Guardrail. | |

Correct answer: B. Contextual Guardrail.

QUESTION 48

A Generative AI Engineer is building a Databricks-hosted assistant that must (1) query Unity Catalog tables with row/column permissions enforced, and (2) avoid managing any external infrastructure. The team wants the LLM to use governed data access through tools exposed via MCP.

Which MCP server choice meets these constraints?

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| A. | Run a custom MCP server on a self-managed VM that proxies the Unity Catalog API. | |
| B. | Use an external community MCP server for SQL and pass Unity Catalog tokens in prompts. | |
| C. | Use a managed Databricks MCP server integrated with Unity Catalog. | CORRECT |
| D. | Expose JDBC directly to the model and enforce permissions in application code. | |

Correct answer: C. Use a managed Databricks MCP server integrated with Unity Catalog.

QUESTION 49

Which indicator should be considered to evaluate the safety of the LLM outputs when qualitatively assessing LLM responses for a translation use case?

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|----|---|----------------|
| A. | The similarity to the previous language. | |
| B. | The latency of the response and the length of text generated. | |
| C. | The ability to generate responses in code. | |
| D. | The accuracy and relevance of the responses. | CORRECT |

Correct answer: D. The accuracy and relevance of the responses.

QUESTION 50

A Generative AI Engineer is building a RAG application that will rely on context retrieved from source documents that are currently in PDF format. These PDFs can contain both text and images. They want to develop a solution using the least amount of lines of code.

Which Python package should be used to extract the text from the source documents?

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|----|---------------|---------|
| A. | numpy | |
| B. | unstructured | CORRECT |
| C. | flask | |
| D. | beautifulsoup | |

Correct answer: B. unstructured

QUESTION 51

A Generative AI Engineer is using the code below to test setting up a vector store:

```
from databricks.vector_search.client import VectorSearchClient

vsc = VectorSearchClient()

vsc.create_endpoint(
    name='vector_search_test',
    endpoint_type='STANDARD'
)
```

Assuming they intend to use Databricks managed embeddings with the default embedding model, what should be the next logical function call?

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|----|--------------------------------------|----------------|
| A. | vsc.similarity_search() | |
| B. | vsc.get_index() | |
| C. | vsc.create_delta_sync_index() | CORRECT |
| D. | vsc.create_direct_access_index() | |

Correct answer: C. vsc.create_delta_sync_index()

QUESTION 52

A Generative AI Engineer wants to build an LLM-based solution to help a restaurant improve its online customer experience with bookings by automatically handling common customer inquiries. The goal of the solution is to minimize escalations to human intervention and phone calls while maintaining a personalized interaction. To design the solution, the Generative AI Engineer needs to define the input data to the LLM and the task it should perform.

Which input/output pair will support their goal?

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|----|--|----------------|
| A. | Input: Online chat logs; Output: Group the chat logs by users, followed by summarizing each user's interactions. | |
| B. | Input: Customer reviews; Output: Classify review sentiment. | |
| C. | Input: Online chat logs; Output: Buttons that represent choices for booking details. | CORRECT |
| D. | Input: Online chat logs; Output: Cancellation options. | |

Correct answer: C. Input: Online chat logs; Output: Buttons that represent choices for booking details.

QUESTION 53

A Generative AI Engineer received the following business requirements for an external chatbot. The chatbot needs to know what types of questions the user asks and route them to appropriate models to answer the questions. For example, the user might ask about upcoming event details. Another user might ask about purchasing tickets for a particular event.

What is an ideal workflow for such a chatbot?

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|-----------|---|----------------|
| A. | The chatbot should only process payments. | |
| B. | There should be two different chatbots handling different types of user queries. | |
| C. | The chatbot should only look at previous event information. | |
| D. | The chatbot should be implemented as a multi-step LLM workflow. First, identify the type of question asked, then route the question to the appropriate model. If it is an upcoming event question, send the query to a text-to-SQL model. If it is about ticket purchasing, the customer should be redirected to a payment platform. | CORRECT |

Correct answer: D. The chatbot should be implemented as a multi-step LLM workflow. First, identify the type of question asked, then route the question to the appropriate model. If it is an upcoming event question, send the query to a text-to-SQL model. If it is about ticket purchasing, the customer should be redirected to a payment platform.

QUESTION 54

A Generative AI Engineer is responsible for developing a chatbot to enable their company's internal HelpDesk Call Center team to more quickly find related tickets and provide resolution. While creating the GenAI application work breakdown tasks for this project, they realize they need to start planning which data sources (either Unity Catalog volume or Delta table) they could choose for this application. They have collected several candidate data sources for consideration:

- **call_rep_history**: a Delta table with primary keys representative_id and call_id. This table is maintained to calculate representatives' call resolution from fields call_duration and call_start_time.
- **transcript Volume**: a Unity Catalog Volume of all recordings as .wav files, but also a text transcript as .txt files.
- **call_cust_history**: a Delta table with primary keys customer_id and call_id. This table is maintained to calculate how much internal customers use the HelpDesk to make sure that the charge back model is consistent with actual service use.
- **call_detail**: a Delta table that includes a snapshot of all call details updated hourly. It includes root_cause and resolution fields, but those fields may be empty for calls that are still active.
- **maintenance_schedule**: a Delta table that includes a listing of both HelpDesk application outages as well as planned upcoming maintenance downtimes.

They need sources that could add context to best identify ticket root cause and resolution. Which TWO sources do that?

A.	transcript Volume	CORRECT
B.	call_rep_history	
C.	maintenance_schedule	
D.	call_detail	CORRECT
E.	call_cust_history	

Correct answer: A. transcript Volume; D. call_detail

QUESTION 55

A Generative AI Engineer developed an LLM application using the provisioned throughput Foundation Model API. Now that the application is ready to be deployed, they realize that the volume of requests is not sufficiently high enough to create their own provisioned throughput endpoint. They want to choose a strategy that ensures the best cost-effectiveness for their application.

What strategy should the Generative AI Engineer use?

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|-----------|--|----------------|
| A. | Deploy the model using pay-per-token throughput as it comes with cost guarantees. | CORRECT |
| B. | Change to a model with a fewer number of parameters in order to reduce hardware constraint issues. | |
| C. | Throttle the incoming batch of requests manually to avoid rate limiting issues. | |
| D. | Switch to using External Models instead. | |

Correct answer: A. Deploy the model using pay-per-token throughput as it comes with cost guarantees.

QUESTION 56

A Generative AI Engineer has deployed a RAG application to production. Its Vector Search index is built from a Delta table that receives incremental updates every hour from an upstream ETL pipeline. The team wants the index to automatically reflect source table changes without manual re-indexing and without changing the endpoint used by downstream applications.

Which approach should the engineer use?

A.	Use a Delta Sync index so changes in the source Delta table are automatically synchronized to the index.	CORRECT
B.	Schedule a job to rebuild the index every hour after the ETL pipeline finishes.	
C.	Replace the current index with a Direct Vector Access index and push updates through a custom streaming job.	
D.	Update the pipeline mode to continuous mode, rather than triggered mode.	

Correct answer: A. Use a Delta Sync index so changes in the source Delta table are automatically synchronized to the index.